

Real-Time Train Tracking & Delay Prediction

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Railway transportation in Sri Lanka serves thousands of passengers daily and remains one of the most affordable and environmentally sustainable modes of public transportation. However, the current railway system provides limited real-time visibility into train locations and lacks reliable mechanisms for predicting train delays. Passengers often depend on static timetables or delayed announcements, which makes it difficult to plan journeys effectively. Additionally, railway operators face operational challenges in monitoring train movements and managing unexpected disruptions across the railway network.

Existing train monitoring systems in many railway networks primarily rely on fixed infrastructure and manual reporting mechanisms, which provide limited real-time updates and lack predictive capabilities. Traditional delay estimation methods typically depend on deterministic scheduling models that do not adequately capture uncertainties in railway operations. As a result, these systems often fail to provide accurate predictions of train arrival times when unexpected operational conditions occur.

The purpose of this study is to propose an Intelligent Real-Time Train Tracking and Probabilistic Delay Prediction Framework that integrates real-time data collection technologies with advanced predictive analytics. The proposed system will utilize technologies such as GPS-based tracking and IoT-enabled data acquisition to continuously monitor train locations and operational parameters. The collected data will be processed using machine learning models and probabilistic prediction techniques to forecast potential delays by analysing historical patterns and real-time operational conditions.

The framework will also incorporate probabilistic delay estimation models capable of capturing uncertainties in train travel times, station dwell times, and network congestion. By analysing multiple influencing factors, the system will generate more accurate delay predictions compared to traditional scheduling-based methods. Furthermore, the proposed system will provide real-time train tracking information and delay predictions through an interactive digital interface, enabling passengers and railway authorities to make informed decisions regarding travel planning and operational management.

The performance of the proposed framework will be evaluated using prediction accuracy metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and prediction reliability under different operational scenarios. Additionally, the system will be analysed in terms of its ability to improve railway service reliability and passenger satisfaction through timely and accurate delay predictions.

The proposed framework aims to transform conventional railway monitoring systems into an intelligent, data-driven platform that enhances real-time operational awareness and predictive decision-making within railway transportation systems.

Keywords: Train Tracking, Delay Prediction, Probabilistic Modelling, Real-Time Railway Monitoring, Machine Learning, Intelligent Transportation Systems.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
SVM	Support Vector Machine
OCSVM	One-Class Support Vector Machine
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
F1-score	Harmonic Mean of Precision and Recall
IoT	Internet of Things
MQTT	Message Queuing Telemetry Transport
GPS	Global Positioning System
SDG	Sustainable Development Goal
WBS	Work Breakdown Structure
UI	User Interface
IDE	Integrated Development Environment
CPU	Central Processing Unit
RAM	Random Access Memory
SSD	Solid State Drive
SST	Software Systems and Technologies
LSTM	Long Short-Term Memory
API	Application Programming Interface
ML Model	Machine Learning Model

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1. INTRODUCTION

1.1 Background and Context

Railway transportation is one of the most widely used modes of public transportation in Sri Lanka. Every day, thousands of commuters rely on railway services for daily travel due to their affordability, accessibility, and efficiency. Railways play a crucial role in connecting urban and rural regions while supporting economic and social activities across the country. Despite its importance, railway transportation systems often face operational challenges that affect the reliability and efficiency of train services. One of the most significant challenges faced by railway transportation is the inability to accurately monitor train movements and predict delays in real time.

Train delays are a frequent issue in many railway systems around the world, including Sri Lanka. Delays may occur due to several factors such as track congestion, signal failures, weather conditions, technical faults, or operational inefficiencies. These delays create inconvenience for passengers who depend on trains for punctual transportation. Passengers often lack access to real-time information about train locations or accurate predictions of arrival times. As a result, passengers experience uncertainty, longer waiting times, and difficulties in planning their journeys effectively.

The consequences of train delays extend beyond passenger inconvenience. Railway operators also experience operational inefficiencies due to delays and the lack of accurate monitoring mechanisms. Without real-time train tracking systems, railway control centres struggle to make timely operational decisions. Delays in one train can cascade into multiple delays across the network, affecting overall service reliability. Additionally, the absence of predictive analytics tools makes it difficult for railway authorities to anticipate delays and take preventive actions before they occur.

Globally, railway transportation systems are increasingly adopting intelligent technologies to improve operational efficiency and passenger experience. Advanced technologies such as Global Positioning Systems (GPS), Internet of Things (IoT), and machine learning-based predictive models are being integrated into railway management systems to monitor train movements and forecast potential delays. Studies indicate that real-time train tracking and predictive delay analysis can significantly improve railway service reliability and passenger satisfaction. However, many developing railway systems still rely on manual monitoring processes and static scheduling systems that cannot adapt to real-time operational changes.

Existing train tracking mechanisms in many railway networks primarily depend on fixed infrastructure such as track circuits, signalling systems, or manual updates from railway staff. These approaches often lack the capability to provide real-time location information and accurate delay predictions. Furthermore, traditional delay estimation methods typically rely on deterministic scheduling models that do not account for uncertainties and probabilistic variations in railway operations. As a result, these systems fail to capture the dynamic nature of train movements and cannot effectively predict delays in complex railway networks.

To address these limitations, this research proposes an Intelligent Real-Time Train Tracking and Probabilistic Delay Prediction Framework. The proposed system integrates real-time train tracking technologies with advanced predictive analytics to provide accurate train location monitoring and delay forecasting. The framework will utilize GPS-based tracking and IoT-enabled data collection mechanisms to continuously monitor train positions and operational parameters. The collected data will then be processed using machine learning algorithms and probabilistic models to estimate potential delays based on historical patterns and real-time operational conditions.

The proposed framework will also incorporate probabilistic delay prediction techniques that consider uncertainties in railway operations. By analysing multiple influencing factors such as travel time variations, station dwell times, track conditions, and congestion levels, the system can generate more accurate delay forecasts compared to traditional deterministic models. In addition, the system will provide real-time updates to passengers and railway authorities through a digital interface, enabling better decision-making and improved travel planning.

The proposed research aligns with several United Nations Sustainable Development Goals (SDGs). In particular, it contributes to SDG 9 – Industry, Innovation and Infrastructure, by introducing intelligent technological solutions that enhance railway infrastructure management. Furthermore, the research supports SDG 11 – Sustainable Cities and Communities, by improving the efficiency and reliability of public transportation systems and enabling smarter mobility solutions for urban populations.

From a research perspective, the proposed work contributes to the Software Systems and Technologies (SST) research cluster, which focuses on developing intelligent and scalable software-based solutions for real-world problems. This research integrates technologies such as real-time data processing, machine learning, probabilistic modelling, and intelligent decision-support systems to develop a smart railway monitoring and prediction platform. By leveraging advanced data analytics and intelligent software systems, the proposed framework aims to enhance railway operational efficiency while improving the overall passenger travel experience.

1.2 Literature Review

The integration of artificial intelligence, Internet of Things (IoT), and data analytics has significantly influenced the development of intelligent transportation systems, particularly in railway operations. Several studies have explored real-time monitoring systems, predictive analytics, and machine learning techniques to improve train tracking, delay prediction, and railway operational efficiency. These studies demonstrate how data-driven technologies can enhance passenger information systems and optimize railway network management.

Research by **Balasingham et al.** proposed a smart railway monitoring architecture that integrates IoT devices and cloud-based communication protocols to support real-time monitoring of railway operations. The system utilizes distributed sensors and the MQTT communication protocol to collect operational data from different railway subsystems. Their study emphasizes the importance of real-time data collection for improving operational visibility and railway management efficiency. However, the proposed architecture mainly focuses on infrastructure monitoring and system integration rather than predicting train delays using probabilistic or machine learning-based models [1].

Similarly, **Silva et al.** introduced the *Smart Rail* system, which applies machine learning techniques to improve railway operational management and passenger information services. The system demonstrates how intelligent algorithms can process large volumes of transportation data to optimize railway services. Although the research highlights the benefits of machine learning for railway management, it primarily focuses on operational monitoring and service optimization rather than providing accurate delay prediction models for railway networks [2].

Another study by **Wanniarachchi et al.** developed a real-time train tracking and arrival time prediction system that utilizes GPS data and analytical models to estimate train arrival times. The system provides passengers with updated train information to improve travel planning. While the proposed system improves passenger information services, it mainly relies on deterministic prediction methods and does not incorporate probabilistic modeling techniques to handle uncertainties in railway operations [3].

Research conducted by **Mihiranga et al.** proposed a machine learning-based railway management system designed to improve the usability and automation of railway operations. The system integrates data processing and predictive analytics to assist railway administrators in decision-making processes. Although the study illustrates how machine learning can support railway operational management, the

proposed solution does not specifically address real-time train tracking or advanced delay prediction mechanisms based on probabilistic models [4].

In addition to train monitoring systems, researchers have explored intelligent technologies for railway infrastructure maintenance and safety monitoring. For example, **Talagune et al.** proposed an automated railway track inspection system using computer vision techniques to detect structural faults in railway tracks. The system demonstrates the effectiveness of artificial intelligence for infrastructure monitoring and railway safety management. However, the focus of this research is on infrastructure inspection rather than real-time train tracking or delay prediction [5].

In the broader context of intelligent transportation systems, **Rathnayaka et al.** developed a machine learning-based traffic violation detection system that analyses vehicle behaviour and predicts traffic rule violations. The study highlights the role of predictive analytics in improving enforcement and transportation management systems. While the research demonstrates how machine learning can support intelligent monitoring systems, it primarily focuses on road traffic management rather than railway operations [6].

Recent research has also investigated predictive analytics approaches for railway delay forecasting. **Singh et al.** introduced the *Rail TechX* framework, which integrates digital technologies and advanced analytics to modernize railway infrastructure and operational management. The study emphasizes the importance of predictive analytics and data-driven decision support systems in improving railway reliability and efficiency. However, the framework mainly focuses on infrastructure modernization rather than developing specialized predictive models for real-time train delay estimation [7].

Further studies in predictive analytics have highlighted the importance of probabilistic modelling techniques for handling uncertainties in transportation systems. Railway operations are inherently uncertain due to various factors such as weather conditions, track congestion, and operational disruptions. Traditional deterministic scheduling models often fail to capture these uncertainties, leading to inaccurate delay predictions. As a result, probabilistic prediction models and machine learning approaches have been increasingly adopted to improve forecasting accuracy in transportation systems [8].

Several machine learning approaches such as regression models, neural networks, and time-series forecasting techniques have been applied to train delay prediction problems. These methods analyze historical train movement data and operational parameters to estimate future delays. Experimental studies show that machine learning-based predictive models can significantly improve delay prediction accuracy compared to traditional timetable-based estimation approaches. Furthermore, combining real-time data with historical patterns enables predictive systems to dynamically adjust predictions based on current operational conditions [9].

Although these studies demonstrate the growing adoption of intelligent technologies in railway monitoring, infrastructure management, and predictive analytics, there remains limited research focusing on **integrated frameworks that combine real-time train tracking with probabilistic delay prediction models**. Particularly in developing railway systems such as Sri Lanka, there is a need for intelligent solutions capable of continuously monitoring train movements while providing reliable delay predictions based on both real-time and historical data. Therefore, this research aims to address this gap by proposing an **Intelligent Real-Time Train Tracking and Probabilistic Delay Prediction Framework** that integrates real-time tracking technologies with machine learning-based predictive analytics to enhance railway operational efficiency and passenger information services.

1.3 Research Gap

Although significant research has been conducted in the field of intelligent transportation systems and railway management, several limitations remain in the existing approaches to train monitoring and delay prediction. Many existing studies focus on improving railway infrastructure monitoring, passenger information services, or general operational management rather than providing a comprehensive solution that integrates real-time train tracking with predictive delay analytics.

Several intelligent railway monitoring systems have been developed using IoT technologies to collect real-time data from railway infrastructure and operational environments. These systems provide valuable insights into railway operations and improve monitoring capabilities. However, most of these solutions mainly focus on infrastructure monitoring and system integration rather than analysing real-time train movement data for predictive delay analysis [1].

Similarly, machine learning-based railway management systems have been proposed to enhance operational decision-making and passenger services. While these systems demonstrate the effectiveness of data-driven technologies in transportation management, they often focus on service optimization and system automation without incorporating advanced predictive models capable of estimating train delays under uncertain operational conditions [2].

Existing train tracking and arrival prediction systems provide real-time location information and estimated arrival times based on historical travel patterns. Although these systems improve passenger awareness and travel planning, many of them rely on deterministic models or simplified statistical approaches that do not adequately capture uncertainties in railway operations such as varying dwell times, track congestion, and unexpected disruptions [3].

Furthermore, many railway delay prediction studies rely heavily on historical datasets and offline analysis methods. Such approaches may fail to provide accurate predictions in dynamic environments where operational conditions change frequently. The absence of real-time data integration and probabilistic modeling techniques limits the accuracy and reliability of delay prediction systems in complex railway networks [4].

Another important limitation in existing research is the lack of integrated frameworks that combine real-time train tracking technologies with probabilistic delay prediction models. Most systems address these aspects separately, either focusing on real-time monitoring or on delay prediction models based solely on historical data. The absence of integrated intelligent systems prevents railway authorities from effectively utilizing real-time operational data to support predictive decision-making and improve service reliability [5].

Additionally, research focusing on intelligent railway monitoring systems in developing transportation networks, particularly in countries such as Sri Lanka, remains limited. Many advanced railway analytics solutions have been developed for highly digitized railway networks in developed countries, where large-scale datasets and advanced infrastructure are available. However, there is a lack of research on scalable and cost-effective predictive railway monitoring frameworks suitable for developing railway systems with limited technological infrastructure [6].

Therefore, there is a clear need for an intelligent framework that integrates real-time train tracking technologies with probabilistic delay prediction models capable of analysing both historical and real-time railway operational data. Such a system would enable railway authorities to monitor train movements more effectively while providing accurate delay predictions that can improve operational planning and passenger information services.

The proposed Intelligent Real-Time Train Tracking and Probabilistic Delay Prediction Framework aims to address these research gaps by developing a unified system that combines GPS-based real-time train tracking, IoT-enabled data acquisition, and machine learning-based probabilistic delay prediction models. By integrating real-time monitoring with predictive analytics, the proposed system seeks to improve railway operational efficiency, enhance passenger information services, and support intelligent decision-making in railway transportation systems.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate an Intelligent Real-Time Train Tracking and Probabilistic Delay Prediction Framework that integrates real-time train location monitoring with machine learning-based predictive analytics in order to accurately track train movements and forecast potential delays within railway networks. The proposed system will analyse both historical and real-time operational data to generate reliable delay predictions and improve passenger information services while supporting operational decision-making within railway management systems.

2.2 Specific Objectives

1. Development of a Real-Time Train Tracking Data Acquisition Framework

The goal of this objective is to design a data acquisition framework capable of collecting real-time train movement data using technologies such as GPS tracking and IoT-enabled communication mechanisms. The framework will capture train location data, travel speeds, station arrival and departure times, and other operational parameters necessary for monitoring train movements in real time.

2. Dataset Preparation and Data Integration for Railway Operations

The objective of this goal is to prepare and integrate railway operational datasets that combine historical train movement data with real-time tracking data. The dataset will include information such as train schedules, station dwell times, route information, historical delay records, and operational conditions. Data preprocessing techniques will be applied to ensure data consistency, completeness, and suitability for predictive analytics.

3. Feature Engineering for Train Movement and Delay Analysis

This objective focuses on extracting and constructing relevant features from the collected railway datasets to represent patterns in train movements and operational conditions. The features may include temporal features such as departure time and travel duration, spatial features such as route segments and station distances, and operational features such as congestion levels, dwell time variations, and historical delay trends. These features will be used to capture patterns that influence train delays.

4. Implementation of Machine Learning and Probabilistic Delay Prediction Models

This objective emphasizes the development and implementation of predictive models capable of forecasting train delays using machine learning and probabilistic modelling techniques. Algorithms such as regression models, time-series forecasting methods, and neural network-based approaches will be explored to predict train arrival times and delay probabilities based on both historical and real-time operational data.

5. Development of a Real-Time Monitoring and Passenger Information System

The goal of this objective is to develop a real-time monitoring platform that visualizes train locations and predicted arrival times through an interactive interface. The system will provide

passengers and railway authorities with real-time updates regarding train positions and expected delays, enabling improved travel planning and operational management.

3. METHODOLOGY

3.1 Research Approach

This research will follow an experimental and data-driven approach for designing, implementing, and evaluating an intelligent framework for real-time train tracking and probabilistic delay prediction in railway transportation systems. This approach is appropriate for the proposed study as it involves the development of predictive models, experimentation with railway operational datasets, and the evaluation of model performance using quantitative performance metrics.

The research methodology begins with the collection and preparation of railway operational datasets, including historical train schedules, station arrival and departure times, route information, and historical delay records. In addition to historical data, real-time train location information will be collected using GPS-based tracking systems and IoT-enabled communication technologies. These datasets will provide comprehensive information regarding train movements and operational conditions within the railway network.

Following the data collection stage, data preprocessing and dataset preparation will be performed. This process includes data cleaning, handling missing values, normalizing numerical values, and integrating multiple data sources into a unified dataset suitable for predictive analytics. The dataset will contain both historical and real-time operational data to ensure that predictive models can capture both long-term patterns and dynamic operational conditions.

After dataset preparation, feature engineering techniques will be applied to extract meaningful features related to train movements and delay patterns. These features may include temporal features such as departure times and travel durations, spatial features such as route distances and station sequences, and operational features such as station dwell times, congestion levels, and historical delay trends. Feature engineering plays a critical role in improving model performance by enabling predictive algorithms to identify patterns associated with train delays.

Once the dataset and feature set are prepared, machine learning and probabilistic models will be trained to predict train delays and estimate expected arrival times. These models will analyse historical train movement data along with real-time operational information to generate delay predictions. Experimental analysis will be conducted to evaluate the effectiveness of these models under different railway operational scenarios.

The final stage of the methodology involves the development of a real-time monitoring and prediction system that integrates train tracking data with delay prediction models. The system will continuously update train locations and provide estimated arrival times and delay probabilities through a visualization interface designed for passengers and railway administrators.

3.2 Tools and Platforms with Justification

The development and experimentation of the proposed train tracking and delay prediction framework will be carried out using widely used tools for data science, machine learning, and real-time data processing.

For the purposes of this research, the primary programming language used will be Python, as it is widely adopted for machine learning research and data analysis. Python provides powerful libraries for data preprocessing, predictive model development, and system integration.

With the aid of Python, the research will efficiently perform data preprocessing, feature engineering, and model development. Python supports several machine learning frameworks and libraries that facilitate the implementation of predictive models and data analysis.

Machine learning libraries available in Python will be used to implement the Linear Regression, Random Forest, and LSTM models, as these libraries provide efficient tools for training and evaluating predictive models. These tools allow the researcher to experiment with different machine learning approaches and compare their performance in predicting train delays.

Additionally, data analysis and visualization tools will be used for preprocessing datasets, exploring train movement patterns, and visualizing prediction results. Visualization tools will help illustrate train locations, predicted arrival times, and delay trends in the railway network.

The use of these tools ensures the efficient implementation of predictive models, reproducibility of experimental results, and scalability of the system for analysing large railway operational datasets.

3.3 Data Collection and Ethical Considerations

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4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 System Overview Diagram

System Overview

The proposed system, named AI-Driven Intelligent Railway Operations System, is intended to assist railway operators, passengers, and transportation authorities by employing artificial intelligence, machine learning, and data analytics technologies to enhance the efficiency of railway operations and the overall experience for passengers. The railway transportation sector in Sri Lanka is also facing some challenges in accurately estimating the demand for passengers, unexpected delays in trains, seat allocation issues, ticket fraud, and static scheduling for trains. The proposed system will address these challenges by developing an intelligent system that analyses the data for railway operations and offers predictive insights for better management in the sector.

The overall system comprises four intelligent components that monitor and optimize different aspects of railway operations.

The Passenger Demand Forecasting and Adaptive Seat Allocation Component uses historical ticketing information, seasonality, and travel trends to forecast the demand for passengers on different routes within a railway system. Using machine learning algorithms, the system predicts the number of passengers expected on different routes within a given time frame. This prediction is then used by the system to allocate seats for passengers, thereby optimizing seat allocation and utilization.

The Real-Time Train Tracking and Delay Prediction Component uses GPS, timestamps, and timetable information to track trains and estimate their delays. This component uses machine learning algorithms and sequence prediction models to estimate the real-time delays of trains, thus providing accurate information on train delays and estimated times of arrival for passengers and operators within a railway system.

The Ticket Fraud Detection and Risk Analysis Component tracks ticketing transactions and passenger pattern behaviours to identify suspicious activity such as ticket reuse, abnormal concession usage, and unauthorized ticket resale. By applying anomaly detection algorithms and machine learning models, the system identifies abnormal ticketing behaviours. This enables ticket inspectors to identify areas to focus on to minimize revenue loss due to fraudulent activity.

The Train Scheduling Optimization and Capacity Management Component uses passenger demand forecasts, delay patterns, and operational constraints to generate optimized train schedules. By applying multi-objective optimization algorithms, the system evaluates alternative train schedules to identify efficient schedules. This enables railway planners to make informed train scheduling decisions.

In general terms, the AI-Driven Intelligent Railway Operations System is a digital platform that combines all these intelligent systems in a way that helps railway authorities monitor passenger demand, predict delays, prevent ticket fraud, and optimize scheduling. The idea behind this system is that by using data from transportation systems along with artificial intelligence technology, railway reliability can be improved, passenger satisfaction increased, and operational inefficiencies reduced .



Figure 1: System Overview Diagram

Figure 1: System overview diagram

4.2 Component Overview

The proposed system provides a framework for an Intelligent Real-Time Train Tracking and Probabilistic Delay Prediction System that is capable of monitoring train movements and predicting potential delays within railway transportation networks. The system utilizes real-time data collection technologies and machine learning-based predictive analytics to analyse train movement patterns and estimate expected arrival times. The main objective of the system is to improve railway operational efficiency and provide passengers with accurate real-time information regarding train locations and possible delays.

The proposed system comprises several components that interact with each other for data collection, processing, analysis, and prediction generation. The system begins with the collection of railway operational data from multiple sources. This data may include real-time train location data obtained through GPS tracking systems, historical train schedules, station arrival and departure records, and route information from railway management systems. These datasets provide comprehensive information regarding train movements and operational conditions within the railway network.

After data collection, the data is pre-processed and transformed into a suitable format for analysis. This process includes cleaning the dataset, handling missing values, normalizing numerical data, and integrating multiple data sources into a unified dataset. Data preprocessing ensures that the dataset is reliable and suitable for machine learning model training and prediction tasks.

Subsequently, feature engineering is carried out, where relevant temporal, spatial, and operational features are extracted from the railway operational data. Temporal features may include departure times, travel durations, and time-of-day patterns. Spatial features may include route segments, distances between

stations, and train positions along the railway network. Operational features may include station dwell times, congestion levels, and historical delay patterns. These features represent the underlying patterns of train movements and operational conditions that influence delay occurrences.

After feature extraction, machine learning algorithms are applied to analyse the pre-processed data and generate delay predictions. Predictive models such as Linear Regression, Random Forest Regression, and Long Short-Term Memory (LSTM) neural networks will be used to estimate train arrival times and predict potential delays. These models analyse historical train movement patterns along with real-time operational data to forecast delay probabilities and expected arrival times for trains.

Once predictions are generated, the system will produce real-time delay forecasts and estimated arrival times for each train within the railway network. These predictions will be displayed through a monitoring interface that visualizes train locations, predicted arrival times, and potential delays. The interface will provide railway administrators with operational insights while enabling passengers to access accurate train tracking and delay information.

The final output of the system includes real-time train location updates, predicted arrival times, and probabilistic delay estimates. These outputs assist railway authorities in monitoring train operations and making informed operational decisions, while also helping passengers plan their journeys more effectively based on reliable real-time information.

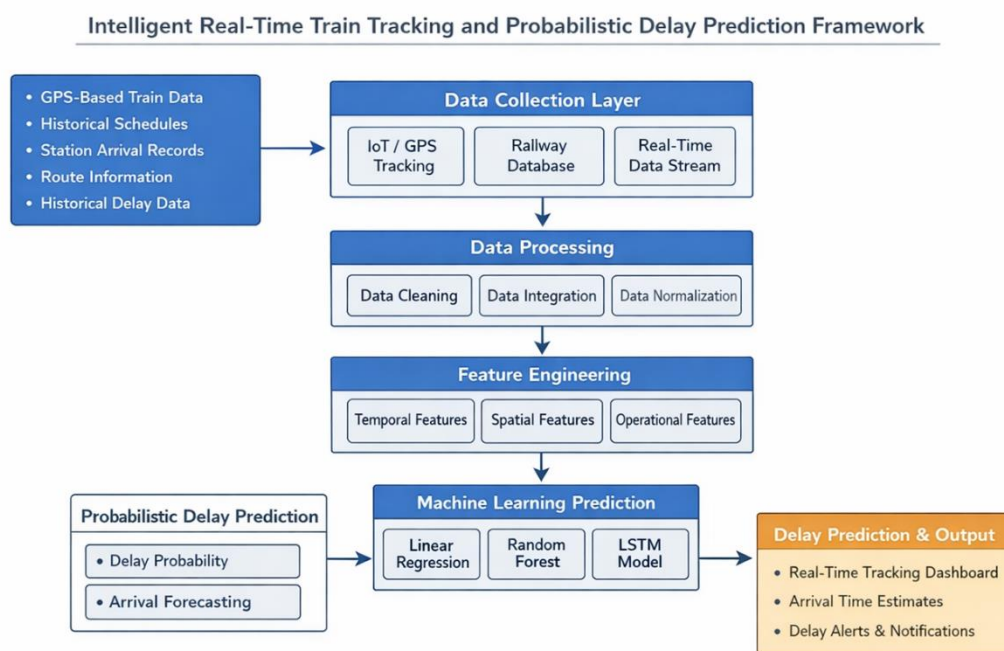


Figure 2: Component overview diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed system involves several stakeholders who interact directly or indirectly with the Intelligent Real-Time Train Tracking and Probabilistic Delay Prediction Framework.

Railway Authorities / Railway Management

The railway authorities are the key stakeholders that manage the railway services effectively by ensuring efficient ticketing services. The authorities need a tool that helps identify fraudulent activities in railway ticketing services. This proposed system helps the railway authorities by providing insights into railway ticketing services and improving the efficiency of ticket inspection services. Train Operation Controllers Train operation controllers are responsible for monitoring train movements and coordinating railway operations from control centres. The proposed framework assists controllers by providing real-time train location updates and predictive delay information. This allows controllers to take proactive actions such as adjusting train schedules, managing track congestion, and coordinating train operations more effectively. Railway Passengers

Railway passengers are indirect users of the proposed system. Passengers rely on railway services for daily transportation and require accurate information about train arrival times and delays. The proposed system benefits passengers by providing real-time train tracking and delay predictions, enabling them to plan their journeys more effectively and reduce waiting times at railway stations.

System Administrators and Data Analysts

System administrators and data analysts are responsible for maintaining and monitoring the performance of the train tracking and delay prediction system. They manage the data infrastructure, monitor system performance, and analyse railway operational datasets. These stakeholders ensure that the predictive models operate efficiently and that the system continues to provide accurate train tracking and delay prediction results.

5.2 Functional Requirements

The functional requirements describe the specific operations and services that the system must perform to support real-time train tracking and delay prediction.

Train Location Data Collection

The system shall collect real-time train location data using GPS-based tracking systems or railway operational data sources.

Railway Operational Data Processing

The system shall process railway operational data such as train schedules, station arrival and departure times, route information, and historical delay records.

Data Preprocessing and Integration

The system shall preprocess and integrate multiple data sources, including real-time train tracking data and historical operational datasets, into a unified dataset for analysis.

Feature Engineering for Delay Analysis

The system shall perform feature engineering to extract relevant temporal, spatial, and operational features that influence train delay patterns.

Delay Prediction Processing

The system shall apply machine learning algorithms such as Linear Regression, Random Forest Regression, and LSTM neural networks to predict train delays and estimate arrival times.

Real-Time Train Monitoring

The system shall display real-time train locations and predicted arrival times through a monitoring interface for railway authorities and passengers.

5.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints of the proposed system.

Security

The system must ensure that railway operational data is protected from unauthorized access. Any sensitive operational data must be handled securely, and access to system data should be restricted to authorized personnel.

Performance

The system should efficiently process large volumes of railway operational data and generate real-time delay predictions within an acceptable time frame.

Usability

The system should provide a user-friendly interface that allows railway authorities and passengers to easily access train tracking information and delay predictions.

Reliability

The system must consistently provide accurate train location updates and delay predictions without interruptions, ensuring reliable service for railway operations and passenger information systems.

Scalability

The system should support the processing of increasing volumes of railway operational data as railway networks expand and additional tracking systems are integrated into the infrastructure. /78569.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The target audience for the proposed system includes railway authorities and transportation organizations responsible for managing railway operations and passenger transportation services. These organizations require intelligent technological solutions that enable them to monitor train operations in real time and improve service reliability through accurate delay predictions.

The primary customers for the proposed system would be national railway authorities, railway management organizations, and transportation infrastructure agencies that manage railway networks. These organizations need advanced analytical systems that provide real-time operational insights and predictive analytics capabilities to improve the efficiency of railway services.

In addition, the system may also be beneficial for urban transportation authorities, smart city transportation planners, and railway infrastructure operators who require intelligent transportation management systems to enhance passenger mobility services. These organizations are interested in adopting innovative technologies that improve railway monitoring, reduce operational disruptions, and enhance passenger satisfaction.

Value Proposition

The proposed system offers a data-driven intelligent solution for monitoring train movements and predicting delays in railway networks. The system integrates real-time train tracking technologies with machine learning-based predictive analytics to provide accurate train location updates and delay predictions.

The value of the proposed system lies in its ability to transform traditional railway monitoring systems into intelligent predictive systems. Instead of relying on static timetables or manual monitoring methods, the system continuously analyses operational data to forecast delays and estimate train arrival times. This enables railway authorities to take proactive actions to minimize service disruptions.

Revenue Model

The proposed solution can be commercialized using multiple revenue strategies.

One possible strategy is offering the system as a licensed software platform that railway authorities can integrate into their existing railway management systems. Under this model, railway organizations would pay a licensing fee to access the predictive analytics platform and real-time train monitoring features.

Another possible revenue strategy is providing the system as a Software-as-a-Service (SaaS) platform. Under this model, railway organizations would subscribe to the service and access train tracking and delay prediction capabilities through a cloud-based platform.

Additionally, the system could generate revenue through data analytics services, where transportation authority's receive analytical insights and reports related to railway operational efficiency and delay patterns.

These revenue models enable transportation organizations to adopt the proposed system without requiring significant internal development efforts.

Cost Estimation

The commercialization of the proposed system involves several cost components, including development, infrastructure, and maintenance costs.

The development cost includes the design and implementation of the real-time train tracking system, machine learning models for delay prediction, and the development of user interfaces for monitoring train operations.

The infrastructure cost includes the cost of servers or cloud-based platforms required for storing railway operational data, processing real-time train location information, and executing predictive models.

The maintenance cost involves continuous system monitoring, software updates, model retraining, and technical support to ensure that the system operates reliably and efficiently.

Pricing Strategy

The pricing model for the proposed system can be designed based on the scale of railway operations and system usage requirements.

Large railway authorities managing extensive railway networks may require more advanced analytics capabilities and higher data processing capacity. These organizations may adopt enterprise-level pricing models that provide comprehensive monitoring and predictive analytics services.

Smaller railway operators or transportation organizations may adopt basic subscription plans that provide essential train tracking and delay prediction functionalities.

The flexible pricing model allows transportation organizations to adopt the system based on their operational requirements and technological capabilities.

Competitive Advantage

The proposed system offers several advantages over existing railway monitoring and scheduling systems.

The first advantage is that the system utilizes machine learning–based predictive analytics to forecast train delays, allowing railway authorities to anticipate operational disruptions before they occur.

The second advantage is the integration of real-time train tracking with delay prediction models, enabling continuous monitoring of train movements and dynamic prediction of arrival times.

Another advantage of the proposed system is its ability to provide probabilistic delay predictions, which consider uncertainties in railway operations such as congestion, dwell time variations, and environmental conditions.

Additionally, the system improves passenger information services by providing accurate real-time train tracking and delay alerts, enhancing the overall passenger travel experience.

Intellectual Property Considerations

The intellectual property associated with the proposed system includes the train tracking framework, predictive delay modeling algorithms, and integrated railway monitoring platform developed in this research. Once the system is commercialized, these components can be protected through software copyrights, licensing agreements, and potential patent protection for the predictive modeling methodologies.

Protecting the intellectual property ensures that the algorithms, software architecture, and predictive analytics techniques developed for the system remain secure and can be commercially deployed by railway authorities while maintaining ownership of the developed technology.

7. BUDGET AND JUSTIFICATION

The development and implementation of the Intelligent Real-Time Train Tracking and Probabilistic Delay Prediction Framework require financial resources for software development, data processing infrastructure, and system maintenance. The estimated budget focuses mainly on software development tools, computational resources, data storage infrastructure, and system deployment costs. The following table summarizes the estimated cost components required for the successful implementation of the proposed system.

Table 1: Budget and justification

Category	Item / Description	Estimated Cost (LKR / USD)	Justification
Hardware Costs	Development Workstation (CPU i7, 16GB RAM, 512GB SSD)	LKR 250,000	Required for processing ticket transaction datasets, feature engineering, and training ML models efficiently.
	Additional Storage (External HDD / SSD 1TB)	LKR 30,000	To store datasets, experimental results, and model checkpoints securely.
Software Licenses	Python IDE (PyCharm Professional / Anaconda)	LKR 20,000	Supports development, debugging, and machine learning model implementation.
	ML Libraries / Tools (scikit-learn, TensorFlow, Py Torch)	Free / Open Source	Necessary for implementing Isolation Forest, One-Class SVM, and Autoencoder models.
Cloud Services	Cloud Storage & Compute (Google Cloud / AWS / Azure)	LKR 50,000	Enables scalable processing of large datasets and simulates real-world railway ticketing environment.
Human Effort Cost	Researcher Development & Testing Effort (200 hours @ LKR 1,500/hr)	LKR 300,000	Covers dataset preparation, feature engineering, model implementation, testing, and documentation.
Total Estimated Cost		LKR 650,000	Total investment required to develop, test, and deploy the intelligent ticket fraud detection system.

Hardware Costs

The hardware requirements for this research project mainly include a development workstation capable of processing railway operational datasets and training machine learning models. The workstation will be used for performing tasks such as data preprocessing, feature engineering, predictive model training, and evaluation of delay prediction algorithms.

In addition, external storage devices will be used to store large railway datasets such as train schedules, route information, and historical delay records. These storage devices will also be used for storing experimental results and trained machine learning models.

The hardware costs are justified since machine learning experimentation and large-scale data analysis require reliable computing resources for efficient system development.

Software Licenses

The software tools used for this project include programming tools and machine learning frameworks required for developing predictive models and analysing railway operational data. Python-based development environments such as PyCharm and Anaconda support efficient coding, debugging, and model implementation.

Machine learning libraries such as scikit-learn, TensorFlow, and Py Torch will be used for implementing predictive models including Linear Regression, Random Forest, and LSTM neural networks for train delay prediction.

These software resources are essential for the research process because they provide the necessary tools for data processing, predictive modelling, experimentation, and performance evaluation.

Cloud Services

Cloud computing environments will be used for storing railway operational datasets and performing large-scale data processing. Cloud platforms such as AWS, Google Cloud, or Microsoft Azure provide scalable computing resources that support machine learning experimentation and real-time data analysis.

Cloud infrastructure also enables the simulation of real-world railway monitoring environments, allowing predictive models to process real-time train tracking data and generate delay forecasts.

The use of cloud services ensures that the system can handle large datasets and support scalable deployment of the predictive analytics framework.

Human Effort Cost

Human effort represents the time and expertise required by the researcher to design, implement, test, and evaluate the proposed system. This includes tasks such as preparing railway operational datasets, developing machine learning models, implementing the real-time train tracking system, conducting experimental evaluations, and documenting the research findings.

Human effort is a critical component of the project since developing machine learning-based predictive systems requires extensive experimentation, data analysis, and algorithm development.

8. WORK BREAKDOWN STRUCTURE (WBS)



Figure 3: Work breakdown structure

9. GANTT CHART

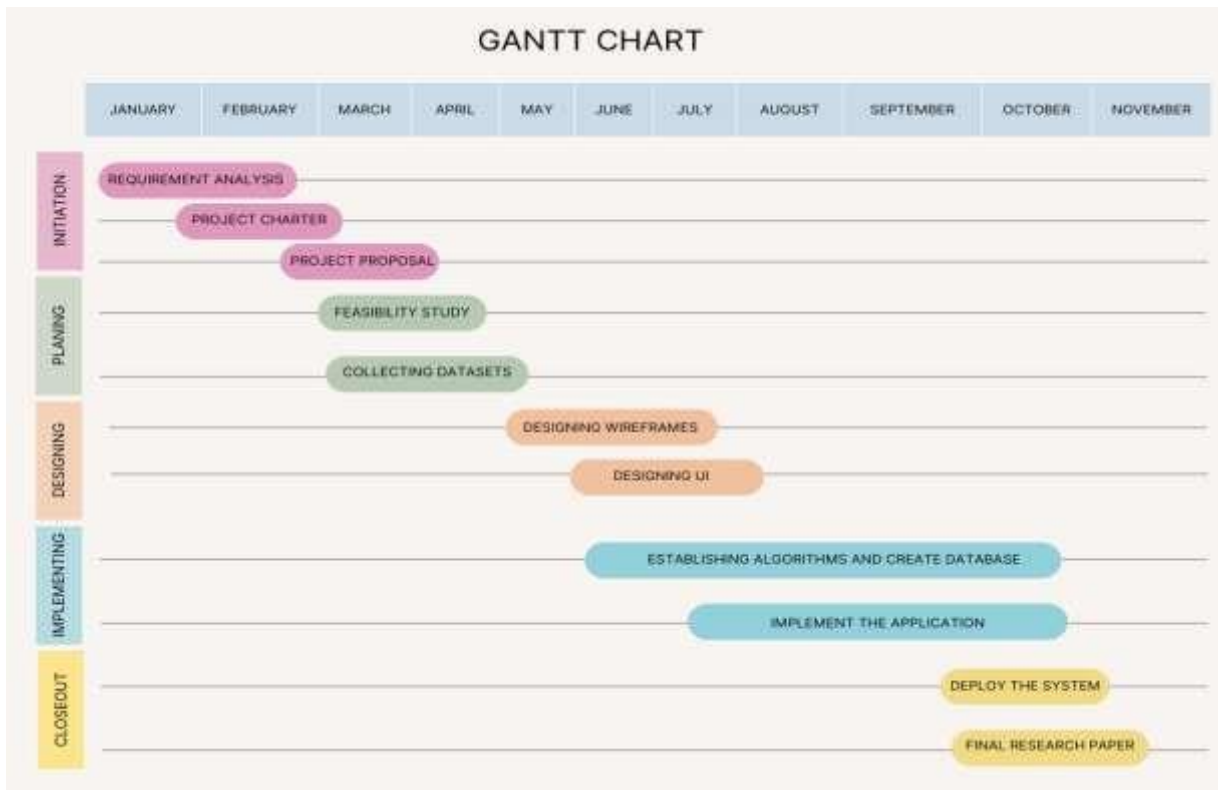


Figure 4: Gantt chart

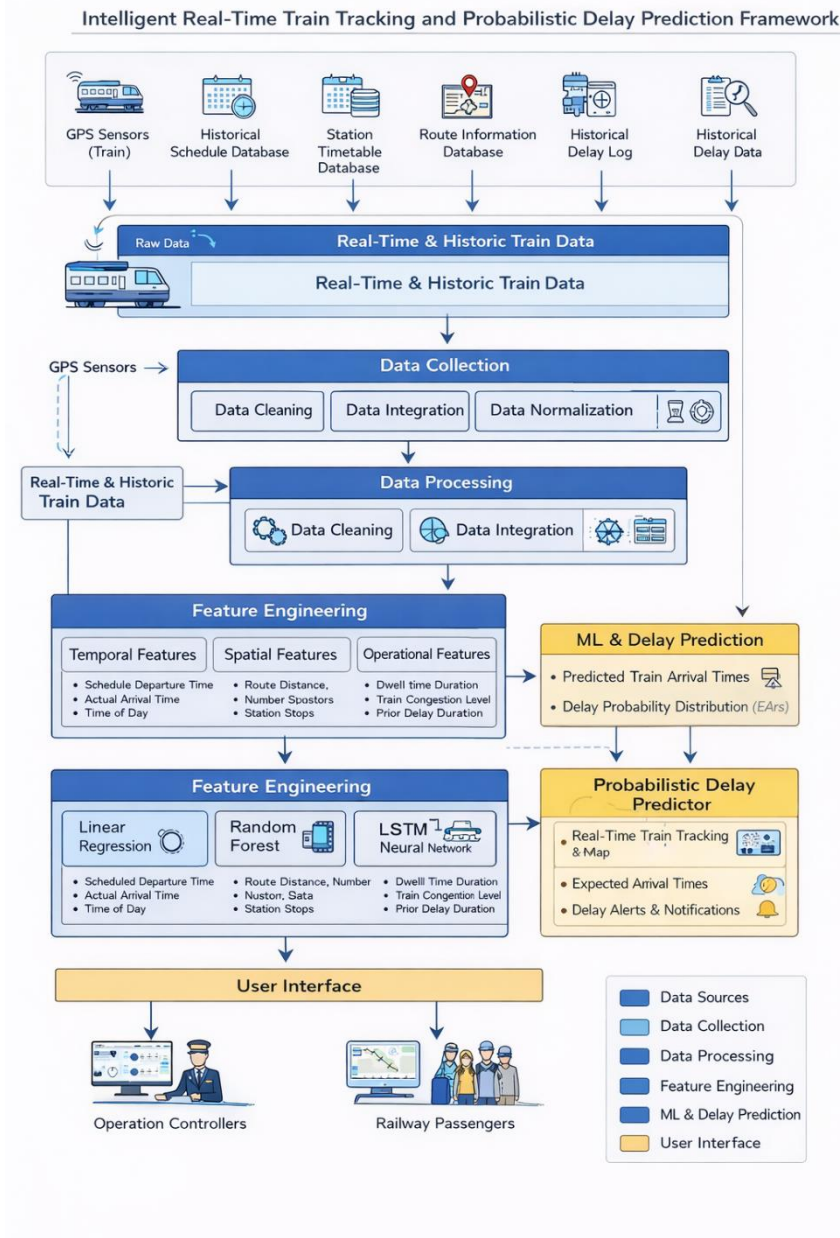
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11.APPENDICES

11.1 Detailed diagrams

Figure 5: Data processing pipeline diag



11.2 Preliminary UI mock-ups

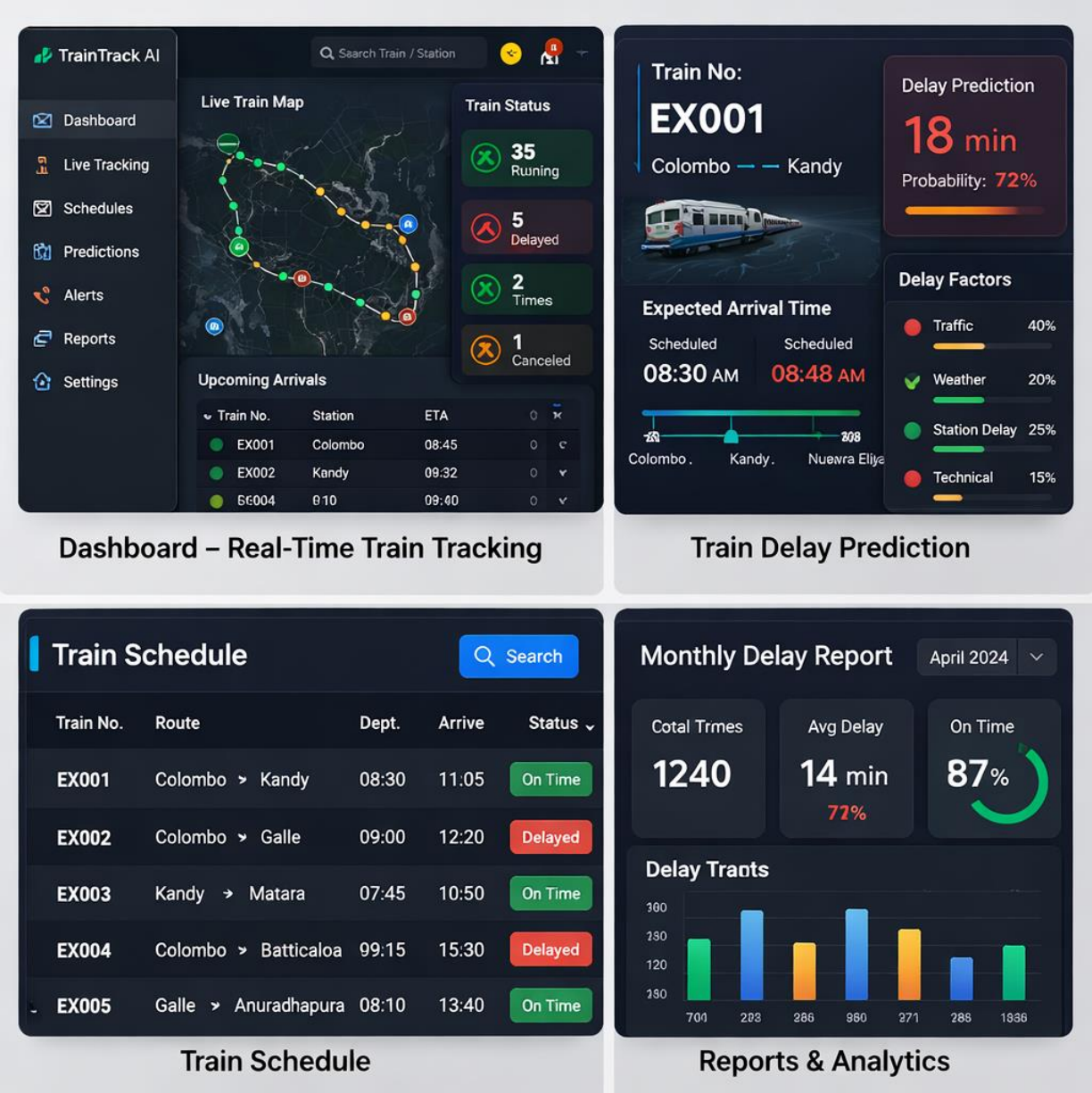


Figure 6: User Interface 1

11.3 Risk analysis

Table 2: Risk analysis

Risk	Impact	Mitigation
Limited availability of real railway operational datasets for training delay prediction models	High	Use publicly available railway datasets and generate simulated datasets that represent realistic train schedules, station dwell times, and delay scenarios.
Incomplete or inaccurate real-time train tracking data from GPS or operational systems	High	Implement data validation mechanisms and filtering techniques to ensure accuracy and reliability of real-time train location data.
Machine learning model accuracy below expected delay prediction performance	High	Apply feature engineering on temporal, spatial, and operational railway data and tune model parameters through iterative experimentation and evaluation.
Real-time processing delays due to large volumes of train movement data	Medium	Optimize data processing pipelines and use scalable cloud computing resources to ensure efficient processing of real-time train tracking data.
Poor quality or incomplete ticket transaction data	Medium	Apply data preprocessing techniques including data cleaning, missing value handling, and normalization before performing anomaly detection analysis.
Difficulty interpreting model outputs by inspectors	Medium	Develop an explanation framework that highlights anomaly severity and key contributing features influencing suspicious ticket transactions.
Data privacy concerns related to passenger ticket records	Medium	Ensure ticket datasets are anonymized and remove personally identifiable information before data processing and model training.
Integration challenges with existing railway management systems	Medium	Design the system architecture using modular components that allow seamless integration with existing railway databases and monitoring systems.
Resistance from railway staff in adopting predictive analytics tools	Low	Resistance from railway staff in adopting predictive analytics tools
Increasing railway network data volume affecting system scalability	Medium	Implement scalable cloud-based infrastructure and optimize machine learning models to handle large volumes of railway operational data efficiently.